A

Mini Project Report

on

LECTURE REMINDER AND TIMETABLE MANANGEMENT SYSYTEM

Submitted in partial fulfillment of the requirements for the degree

Second Year Engineering – Computer Science Engineering (Data Science)

By

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Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on **Lecture Reminder and Timetable Management System** has been submitted by **Umesh Patel** (24107042), **Shrutika Mane** (24107135), **Nupur Sugadare** (24107118) and **Kshitija Patil** (24107092) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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1.

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Place: A. P. Shah Institute of Technology, Thane
Date: 1st April , 2025

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Ms. Swati Parhad** Expressing gratitude towards our HoD, **Dr. Pravin Adivarekar**, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Prof. Richa Singh** and **Prof Kiran Gaikwad** who gave us their valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

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CHAPTER 1

1. Introduction

The proposed mini project, titled "**Lecture Reminder and Timetable Management System**" is designed to facilitate efficient scheduling and communication within an academic environment. The system enables faculty members to create, update, and manage lecture timetables in a structured digital format, allowing students to conveniently access their daily schedules.

In addition to timetable management, the application incorporates a notification mechanism wherein teachers can set reminders for scheduled lectures, including details such as subject, time, and classroom. These notifications are automatically delivered to both students and faculty, ensuring timely awareness and improved coordination.

Furthermore, the system provides a feature for broadcasting important academic notices, such as lecture cancellations or substitutions. This ensures that all students receive real-time updates, thereby minimizing confusion and enhancing overall communication efficiency.

1.1. Purpose

The purpose of the Lecture Reminder and Timetable Management System is to simplify timetable management and ensure timely communication through automated reminders and updates, reducing confusion and missed lectures. This application is intended to help students and faculty members to provide a comprehensive understanding and its practical application in managing academic schedules and communication. It outlines the functioning of the system and highlights how it facilitates efficient timetable access, timely notifications, and the dissemination of important academic updates, thereby enhancing coordination and reducing communication gaps within the academic environment.

1.2. Problem Statement

Current methods of managing and communicating lecture schedules lack efficiency, reliability, and real-time accessibility. This often results in missed lectures, confusion regarding classroom changes, and ineffective communication between teachers and students. Therefore, a centralized mobile-based system is required to automate timetable management, provide instant notifications, and enhance overall academic coordination.

1.3. Objectives

- To provide a centralized and structured system for creating, updating, and sharing lecture schedules between faculty and students.
- To enable real-time notifications and reminders for scheduled lectures, including subject, time, and classroom details.
- To facilitate instant communication of important updates such as lecture cancellations, substitutions, or rescheduling.
- To ensure easy accessibility of timetable and notification information anytime and anywhere through mobile devices.

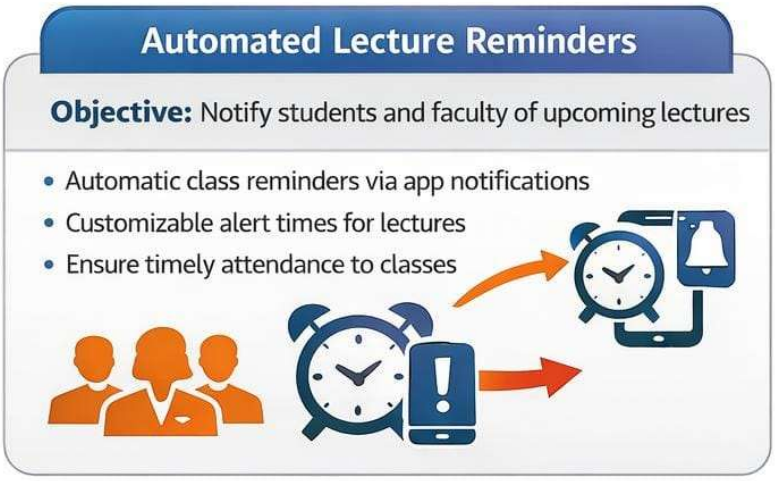

Figure 1.1 : Automated Lecture Reminder

The above figure 1.1 presents the Automated Lecture Reminders system, designed to help students and faculty stay updated about upcoming lectures through automatic app notifications. It allows users to set customizable reminder times, ensuring timely attendance and better class management. The system improves communication, reduces missed lectures, and supports a more organized academic schedule.

Figure 1.2 : Efficient Timetable Management

The above figure 1.2 explains the Efficient Timetable Management system, which helps organize and manage academic schedules in a simple and effective way. It provides easy access to updated timetables, allows faculty to quickly make schedule changes, and ensures centralized management of all class schedules. This system saves time, reduces confusion, and improves overall scheduling efficiency.

1.4. Scope

The proposed system focuses on developing a mobile-based platform for managing lecture timetables, reminders, and academic notifications within an educational institution. It enables faculty to update schedules and send real-time alerts, while allowing students to easily access timetable information and receive timely updates, thereby improving communication and coordination.

CHAPTER 2

2. Proposed System

The proposed system is a mobile application that enables teachers to manage lecture timetables and send real-time notifications. Students can easily access schedules and receive updates regarding lectures, cancellations, and changes, improving communication and coordination within the academic environment.

2.1 Features and Functionality

- Timetable Management
  Faculty members can create, update, and manage lecture timetables by entering details such as subject name, lecture time, and classroom.
- Lecture Schedule Viewing
  Students and teachers can easily access and view the complete timetable anytime through the mobile application.
- Lecture Reminder System
  The system provides automatic notifications before scheduled lectures, ensuring that both students and faculty are reminded in advance.
- Notice Management
  Teachers can send important notices such as lecture cancellations, substitutions, or rescheduling, which are instantly delivered to users.
- Real-Time Notifications
  The application sends real-time alerts regarding lecture timings, updates, and changes to improve communication efficiency.

CHAPTER 3

2. Project Outcomes

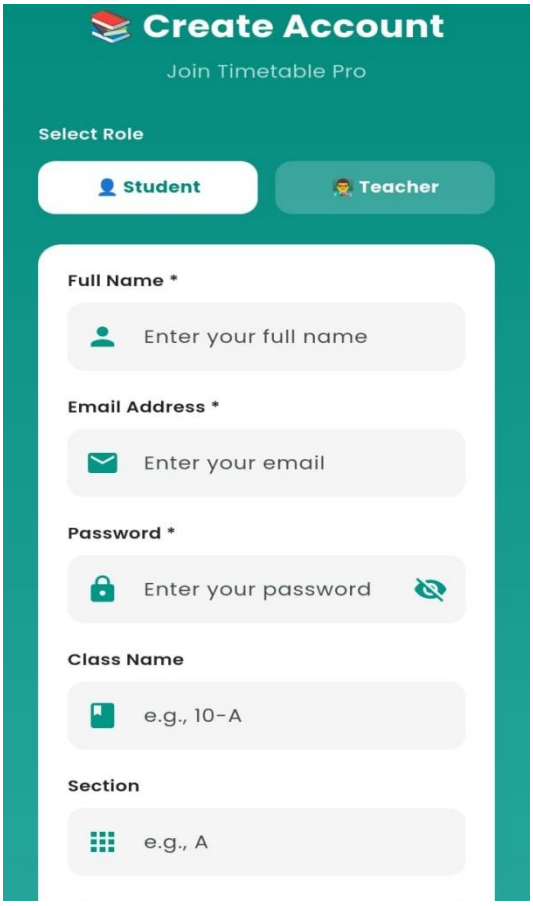

Figure 3.1 Registration Page

This is the Registration Page of the Timetable Pro app, where users can create a new account by selecting their role as Student or Teacher. After choosing the role, users simply enter their basic details such as name, email, password, class name, and section to complete the registration process. It provides a quick and user-friendly way to join the system.

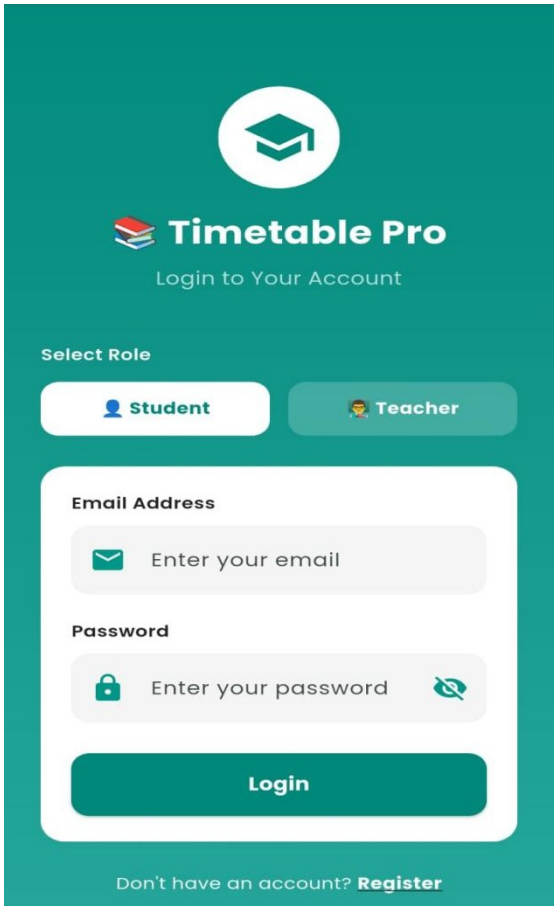

Figure 3.2 Login Page

This is the Login Page of the Timetable Pro app, where users can access their account by selecting their role as Student or Teacher. After choosing the role, they need to enter their email address and password, then click the login button. It provides a simple and secure way for users to enter the system and use its features.

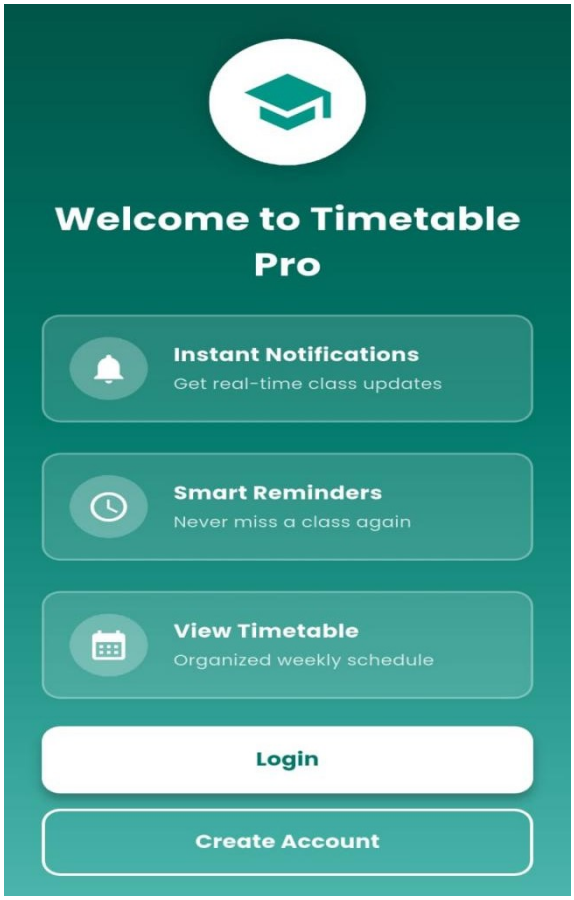

Figure 3.3 Opening Interface

This is the Opening Interface of the Timetable Pro app, which welcomes users and introduces the main features of the system. It highlights instant notifications, smart reminders, and timetable viewing options for better schedule management. From this page, users can easily choose to Login or Create Account to start using the application.

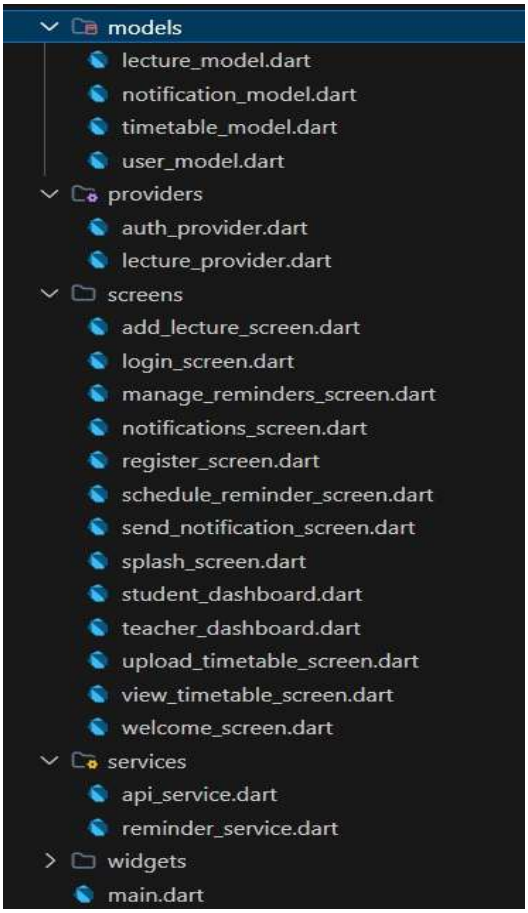

Figure 3.4 Main Code Structure

This image shows the project structure of the Timetable Pro app developed in Flutter. It is organized into folders such as models, providers, screens, and services for better code management. The structure includes files for login, registration, dashboards, timetable viewing, notifications, reminders, and backend services, making the application well-structured, scalable, and easy to maintain.

CHAPTER 4

4. Software Requirements

The development of the proposed system was carried out using modern mobile application technologies to ensure efficiency, usability, and performance. The front-end of the application is developed using **Flutter**, which provides a responsive and visually appealing user interface for both students and teachers.

The backend is implemented using **SQLite**, a lightweight and embedded database that enables local data storage within the application. It is used to manage user information, lecture timetables, and notifications efficiently through structured tables.

The development process involved designing user interfaces, integrating database operations, and implementing core functionalities such as authentication, timetable management, and notification handling. The system was tested to ensure smooth performance, reliability, and ease of use.

CHAPTER 5

5. Project Design

The proposed system is designed using a modular and user-centric approach to ensure simplicity, efficiency, and scalability. The application is divided into two main components: the front-end, developed using Flutter, and the backend, implemented using SQLite for local data storage.

The system consists of key modules including User Authentication, Timetable Management, Reminder/Notification System, and Notice Management. The authentication module ensures secure login and role-based access for students and teachers. The timetable module allows faculty to create and manage lecture schedules, while students can view them in real time.

The notification module is responsible for generating and delivering reminders for scheduled lectures, ensuring that both students and teachers are informed in advance. Additionally, the notice module enables teachers to broadcast important updates such as lecture cancellations or rescheduling.

The interaction between the front-end and backend is handled through structured database operations, ensuring smooth data flow and efficient performance. The overall design focuses on providing a simple, reliable, and user-friendly experience for academic schedule management.

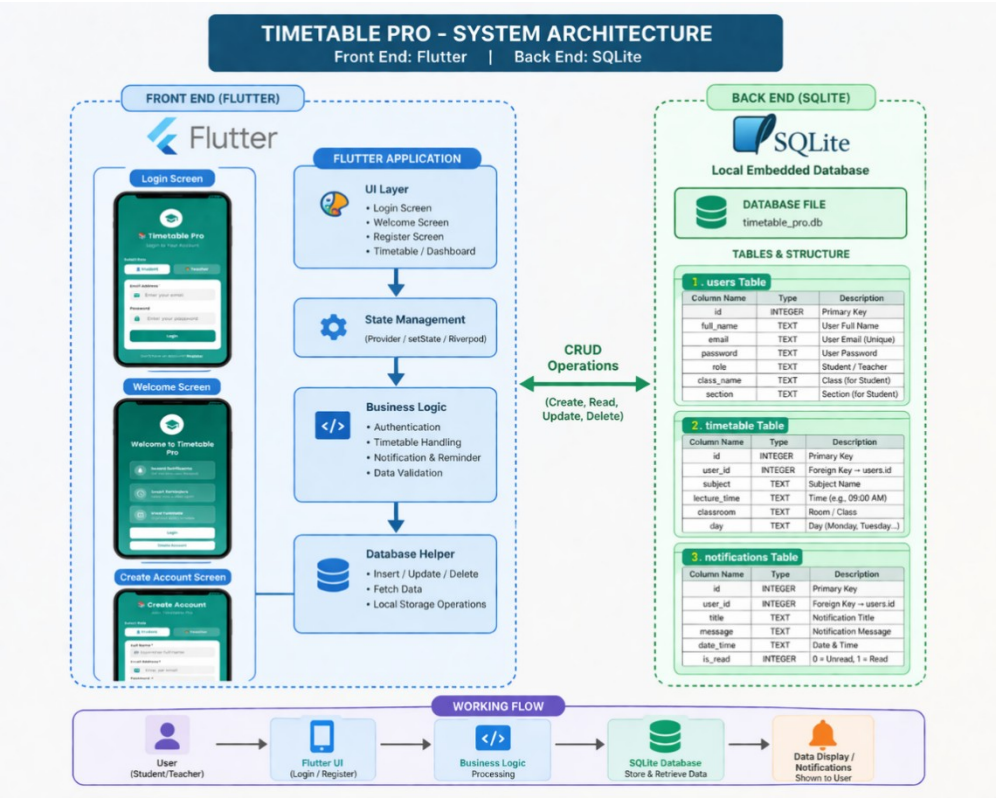

Figure 5.1 System Architecture Diagram

Module Name	Description
Authentication Module	Handles user login and registration with role-based access (Student/Teacher)
Timetable Module	Allows teachers to create, update, and manage lecture schedules
Notification Module	Sends reminders and alerts for scheduled lectures
Notice Module	Enables teachers to post notices like cancellations or changes
Database Module	Stores user data, timetable, notifications, and notices using SQLite

Table 5.1 Module Table

This table shows the main modules of the Timetable Pro system and their functions. It includes the Authentication Module for login and registration, Timetable Module for managing lecture schedules, Notification Module for reminders and alerts, Notice Module for important updates, and Database Module for storing all system data using SQLite. These modules work together to ensure smooth and efficient management of academic activities.

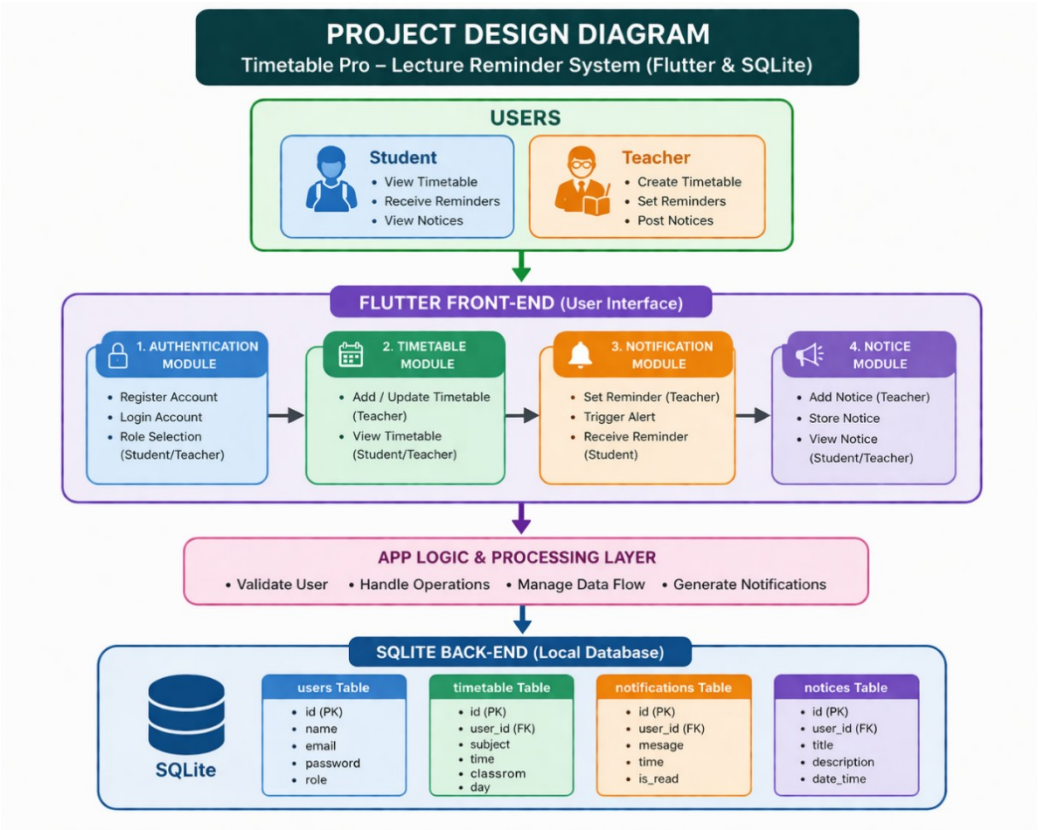

Figure 5.2 Project Design Diagram

This is the Project Design Diagram of the Lecture Reminder System developed using Flutter and SQLite. It shows two types of users: Student and Teacher, where students can view timetables, reminders, and notices, while teachers can create timetables, set reminders, and post notices. The Flutter front-end includes modules such as Authentication, Timetable, Notification, and Notice, while the app logic layer manages operations and data flow. All data is stored in the SQLite database, making the system efficient, organized, and easy to use.

CHAPTER 6

6. Project Scheduling

PROJECT TITLE: Lecture Reminder and Timetable management system

PROJECT GUIDE: MS.SWATI PARHAD

INSTUTUTE & DEPARTMENT NAME: AP SHAH INSTITUTE OF TECHNOLOGY/CSE-Data Science

DATE:1/4/2026

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	PERCENTAGE OF TASK COMPLETE
1	**Project Conception and Initiation**					
1.1	Group formation and Topic finalization. Identifying the scope and objectives of the Mini Project	All	13/1/26	16/1/26	1	100%
1.2	Identifying the functionalities of the Mini Project	All	16/1/26	20/1/26	1	100%
1.3	Discussing the project topic with the help of paper prototype.	All	20/1/26	3/2/26	2	100%
1.4	Designing the Graphical User Interface(GUI)	Umesh Patel And Nupur Sugadare	3/2/26	17/2/26	2	100%
1.5	Presentation I	All	17/2/26	24/2/26	1	100%
2	**Project Design and Implementation**					
2.1	Database Design	Kshitija Patil And Shrutika Mane	24/2/26	10/3/26	2	100%
2.2	Database Connectivity of all modules	Umesh Patel And Nupur Sugadare	10/3/26	17/3/26	1	100%
2.3	Integration of all modules and Report Writing	Kshitija Patil And Shrutika Mane	17/3/26	24/3/26	1	100%
2.4	Presentation II	All	24/3/26	1/4/26	1	100%

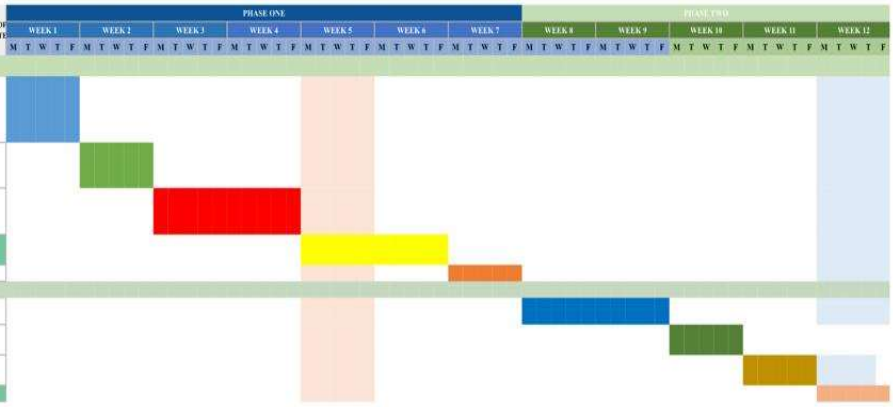

Figure 6.1 Gantt Chart

This Gantt chart presents the project schedule for the Smart Timetable & LectureReminder Management System from January to April 2026. It is divided into threephases: Planning & Design, Development, and Testing & Deployment. Each taskincludes start date, due date, duration, completion percentage, and owner. The chartvisually tracks progress using colored bars, helping monitor task timelines and overall project completion efficiently.

PROJECT WORK DISTRIBUTION

Timetable and Lecture Reminder Management System

Sr No	Activity	Phase	% Done	Assigned To
1	Requirement Analysis	Planning & Design	100%	Kshitija Patil
2	System Design (ER Diagram)	Planning & Design	100%	Kshitija Patil
3	UI/UX Design (App Screens)	Planning & Design	90%	Shrutika Mane
4	Database Setup & Configuration	Development	100%	Umesh Patel
5	Authentication Module (Login/Register Programming)	Development	95%	Nupur Sugadare
6	Timetable Management Module Programming	Development	85%	Umesh Patel
7	Lecture Reminder & Notification Module Programming	Development	70%	Nupur Sugadare
8	Notice Management (Cancel/Replace Lecture)	Development	60%	Shrutika Mane
9	Testing	Testing & Deployment	—	Shrutika Mane
10	Documentation	Testing & Deployment	100%	Kshitija Patil
11	Bug Fixing	Testing & Deployment	—	Nupur Sugadare
12	Deployment	Testing & Deployment	100%	Umesh Patel
13	Final Integration	Testing & Deployment	90%	All

Table 6.2 Project Work Distribution

This table shows the project work distribution among team members for the Timetable and Lecture Reminder Management System. The tasks included planning, documentation, system design, UI/UX designing, programming, development modules, testing, and final integration. The work was divided properly among all members to ensure smooth teamwork, efficient coordination, and successful project completion .

CHAPTER 7

7. Results

The developed system was successfully implemented using Flutter and SQLite, providing a smooth and user-friendly interface for both students and teachers. The application allows teachers to efficiently create and manage lecture timetables, set reminders, and post important notices.

Students are able to view their schedules and receive real-time notifications for upcoming lectures, cancellations, and updates. The system ensures accurate data storage and quick retrieval through SQLite, resulting in reliable performance and reduced chances of missed lectures.

Overall,the application demonstrated effective communication, improved coordination, and enhanced time management within the academic environment.

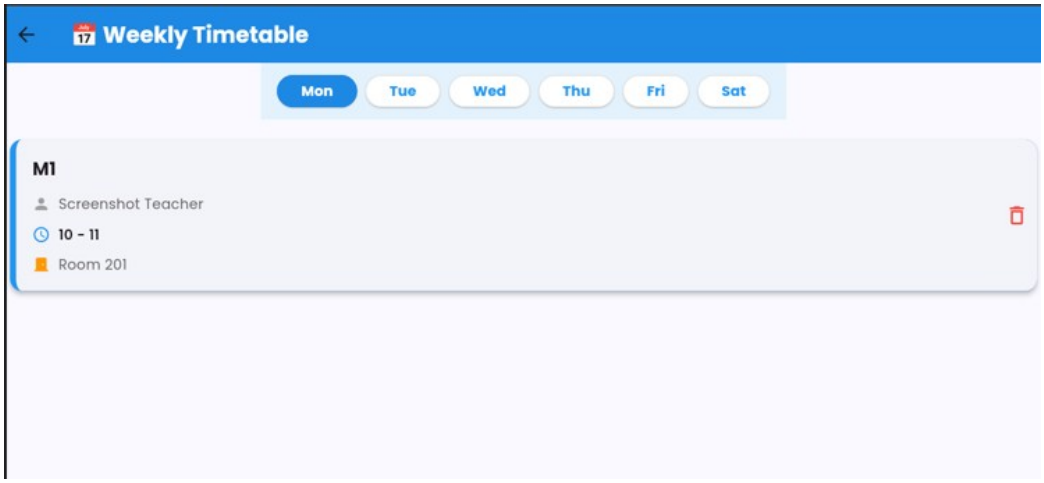

Figure 7.1 Timetable

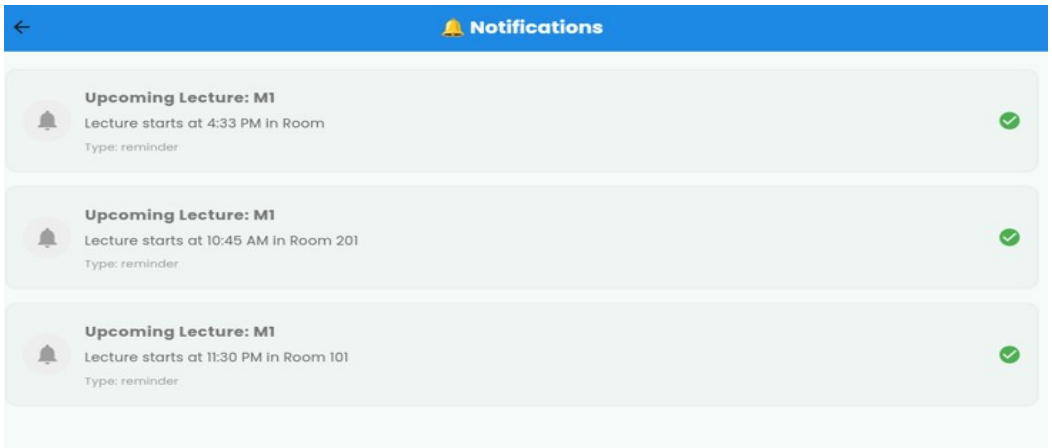

Figure 7.2 Notification on Student and Teacher Dashboards

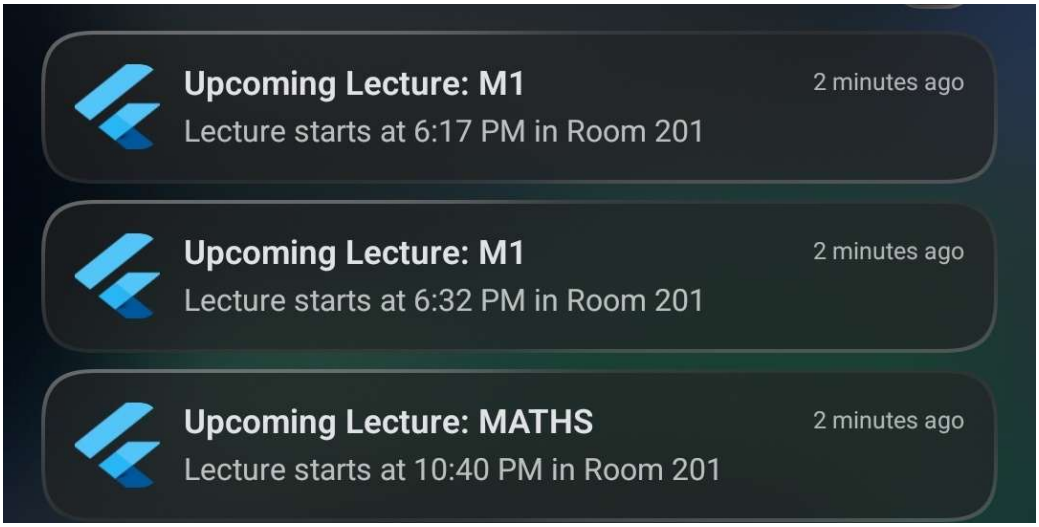

Figure 7.3 Notifications for Student and Teacher

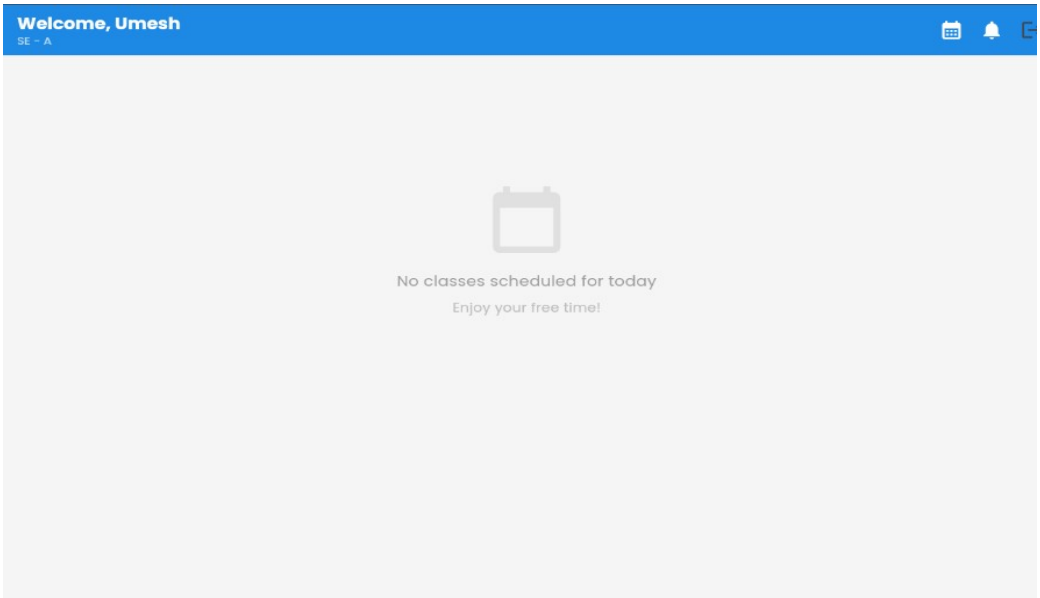

Figure 7.4 Student Dashboard

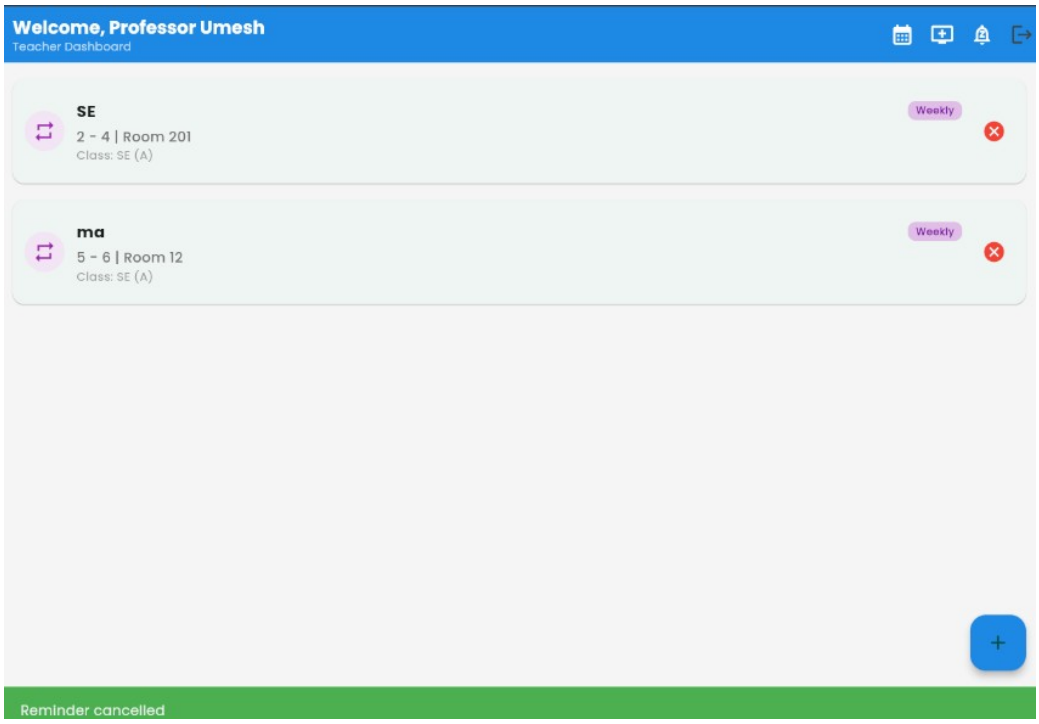

Figure 7.5 Teacher Dashboard

These Figures displays the main interfaces of the Timetable and Lecture Reminder Management System, including timetable view, notifications, student dashboard, and teacher dashboard. The system allows users to check weekly lecture schedules, receive upcoming lecture reminders, and manage daily academic activities efficiently. Students can view schedules and notifications, while teachers can update lectures and monitor timetable information. Overall, the application provides a smart, organized, and user-friendly platform for smooth timetable management

CHAPTER 8

8. Conclusion

The Lecture Reminder and Timetable Management System was successfully designed and developed using Flutter and SQLite, providing an efficient platform for managing academic schedules and notifications. The system enables teachers to create and update lecture timetables, set reminders, and share important notices, while students can easily access schedules and receive real-time updates.

The application enhances communication between teachers and students, reduces the chances of missed lectures, and improves overall time management. With its user-friendly interface and reliable performance, the system serves as a practical solution for modern educational environments.

CHAPTER 9

9. References

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   → Used for frontend development.

5. Firebase Documentation
   https://firebase.google.com/docs
   → Authentication, database, and notifications.

6. SQLite Documentation
   https://www.sqlite.org/docs.html
   → Local data storage for timetable.

7. FlutterFlow
   https://flutterflow.io
   → Used for UI building and backend integration.